Mathematical Foundations for Machine Learning | MathFML

Summary

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1. LINEAR ALGEBRA

Definition 1 : Linear combination

$$\sum_{i=1}^{k} \lambda_i \vec{v}_i$$

1.1. STANDARD BASIS

Basis vectors $\{\vec{e}_1, \vec{e}_2, ..., \vec{e}_n\}$ in $\mathbb{R}^n$ where $\vec{e}_i = \left(0, 0, ..., \underbrace{1}_{i\text{th position}}, ..., 0\right)^T$

Example 1

$$\left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right] = \mathbb{R}^3$$

It can be shown that any basis of $\mathbb{R}^n$ consists of exactly n linear independent vectors and that conversely any set of n linear independent vectors is a basis of $\mathbb{R}^n$.

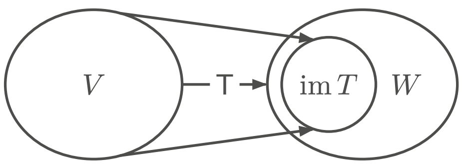
1.2. VECTORS

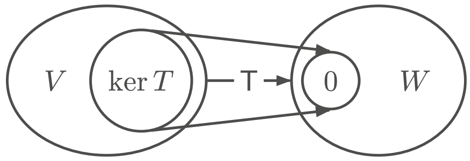
TODO:
group and clean this up

$$\begin{gathered}
\lambda\vec{0} = \vec{0} \\
\vec{v} + \vec{0} = \vec{v} \\
-\vec{v} = -1 \cdot \vec{v} \\
-\vec{v} + \vec{v} = \vec{0} \\
(\lambda\mu)\vec{v} = \lambda(\mu\vec{v}) = \lambda\mu\vec{v} \\
\lambda(\vec{v} + \vec{w}) = \lambda\vec{v} + \lambda\vec{w} \\
\vec{v} + (\vec{u} + \vec{w}) = (\vec{v} + \vec{u}) + \vec{w} = \vec{v} + \vec{u} + \vec{w}
\end{gathered}$$

1.3. LINEAR TRANSFORMATIONS

$$T : V \to W$$

Term	Definition
Image / range	$\operatorname{im} T = \{T(v) \mid v \in V\} = T(V)$ 

Term	Definition
Kernel / nullspace	$\ker T = \{v \in V \mid T(v) = 0\}$ 

Observations 1

1.1. $\operatorname{im} T \subset W$

1.2. $\ker T \subset V$

1.3. $\vec{0} \in \ker A$

Example 2

$$A \in \mathbb{R}^{n \times n}, T_A : \begin{cases} \mathbb{R}^n & \to \mathbb{R}^n \\ \vec{x} & \mapsto A\vec{x} \end{cases}$$

$$\ker A = \left\{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\right\} = \left\{\vec{x} \in \mathbb{R}^n \mid T_A(\vec{x}) = \vec{0}\right\}$$

TODO:

1.3.1. Affine transformations

All linear transformations share the property, that they map the null vector to null:

$$L_A\left(\vec{0}\right) = \vec{0}$$

For many applications this is undesirable. Therefore linear transformations are often slightly extended to transformations, which are called **affine transformations**. An affine transformation is defined by an $r \times c$ matrix M and an r-dimensional vector b, which, in machine learning and statistics, is usually called **bias-vector** or **intercept**.

$$A_{b,M} : \begin{cases} \mathbb{R}^c & \to \mathbb{R}^r \\ x & \mapsto b + M \cdot x \end{cases}$$

Example 3 : Straight line in one dimension

$$A_{b,m} : \begin{cases} \mathbb{R} \to \mathbb{R} \\ x \mapsto mx + b \end{cases}$$

1.4. MATRICES

Term	Definition
Rank	Maximum number of linearly independent rows or columns $$A \in \mathbb{R}^{m \times n} \Rightarrow \operatorname{rank}(A) = \rho(A) \leq n$$ **TODO:** check
Nullity	Number of vectors in the null space $$A \in \mathbb{R}^{m \times n} \Rightarrow \operatorname{nullity}(A) = n - \rho(A) = \dim(\ker A)$$ **TODO:** check
Dimension	$$\dim(\ker A) = \operatorname{nullity}(A)$$ $$\vec{x} \in \mathbb{R}^3 \Rightarrow \dim \vec{x} = 3$$
Span	Set of all finite linear combinations of the elements of S of a vector space V. $$\operatorname{span}(S) = \{\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + ... + \lambda_n \vec{v}_n \mid n \in \mathbb{N}, v_1, ..., v_n \in S, \lambda_1, ..., \lambda_n \in K\}$$ $$\operatorname{span}\{(1,0,0),(0,1,0),(0,0,1)\} = \mathbb{R}^3$$
Column space	Span of the column vectors of a matrix $$\operatorname{colsp}(A) = C(A)$$ $$A \in \mathbb{R}^{m \times n}, \dim(\operatorname{colsp}(A)) = n$$
Row space	Span of the row vectors of a matrix $$\operatorname{rowsp}(A) = C(A^T)$$ $$A \in \mathbb{R}^{m \times n}, \dim(\operatorname{rowsp}(A)) = m$$

***Example** 4*

TODO:

***Observations** 2*

2.1. $\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim V$
2.2. $\dim(\operatorname{im} T) + \dim(\ker T) = \dim(\operatorname{domain} T)$
2.3. $\operatorname{rank}(A) = \dim(\operatorname{rowsp}(A)) = \dim(\operatorname{colsp}(A))$
2.4. $A \in \mathbb{R}^{n \times n} \Rightarrow \dim(\ker A) + \operatorname{rank} A = n$
2.5. $A \in \mathbb{R}^{n \times n}, \ker A = \left\{\vec{0}\right\} \Leftrightarrow \operatorname{rank} A = n$

1.4.1. Properties

Term	*Rules*
Associativity	$(A+B)+C = A+(B+C) = A+B+C$ $(A \cdot B) \cdot C = A \cdot (B \cdot C) = A \cdot B \cdot C$
Distributivity	$C \cdot (A+B) = C \cdot A + C \cdot B$ $(A+B) \cdot C = A \cdot C + B \cdot C$
Commutativity	$A+B = B+A$ $A \cdot B \neq B \cdot A$
Transposing	$\left(A^T\right)^T = A$ $(A+B)^T = A^T + B^T$ $(\lambda A)^T = \lambda A^T$ $(A \cdot B)^T = B^T \cdot A^T$ $A_{ij} = A^T_{ji}$
Identity matrix	$\mathbb{1} \cdot A = A \cdot \mathbb{1} = A \text{ for } A \in \mathbb{R}^{n \times n}$
Inverting	$A \cdot A^{-1} = A^{-1} \cdot A = \mathbb{1}$
Determinate	$\det(\lambda M) = \lambda^n \det(M), M \in \mathbb{R}^{n \times n}$ $\det\begin{pmatrix} A & * \\ 0 & B \end{pmatrix} = \det(A) \cdot \det(B)$ $\det(A \cdot B) = \det(A) \cdot \det(B)$ $\det(A^{-1}) = \frac{1}{\det(A)}$ $\det(A^T) = \det(A)$
Scalars	$(\lambda + \mu)A = \lambda A + \mu A$ $(\lambda \mu)A = \lambda(\mu A)$ $\lambda(B+C) = \lambda B + \lambda C$ $\lambda(BC) = (\lambda B)C = B(\lambda)C$

1.4.2. Unit matrices

Let M be the set of all 2×2 matrices. How could the basis of M look? $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$
Unit matrices:

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$M = \{E_{11}, E_{12}, E_{21}, E_{22}\}$$

$$(E_{11})_{22} = 0, (E_{11})_{11} = 1$$

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \in \mathbb{R}^3, (\vec{e}_1)_1 = 1$$

1.4.3. Kronecker Delta

$$\delta_{ij} = \begin{cases} 1 \text{ , } i = j \\ 0 \text{ , otherwise} \end{cases}$$

***Example** 5*

$$\delta_{23} = 0, \delta_{11} = 1, (\vec{e}_1)_j = \delta_{1j} = \begin{cases} 1 \text{ , } 1 = j \\ 0 \text{ , otherwise} \end{cases}$$

$$\delta_{m1}\delta_{m2} = 0$$
$$\delta_{kl} - \delta_{lk} = 0$$
$$(e_k)_i = \delta_{ki} = \delta_{ik}$$
$$(E_{kl})_{ij} = \delta_{ki}\delta_{lj}$$
$$(E_{rst})_{ijk} = \delta_{ri}\delta_{sj}\delta_{tk}$$

1.4.4. Matrix product

$$A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{m\times r}, A \cdot B \in \mathbb{R}^{n\times r}$$

$$A \cdot B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{im} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1r} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i1} & b_{i2} & \dots & b_{ir} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mr} \end{pmatrix}$$

$$(A \cdot B)_{11} = a_{11}b_{11} + a_{12}b_{12} + \dots + a_{1m}b_{m1} = \sum_{k=1}^{m} a_{1k}b_{k1}$$

$$(A \cdot B)_{ij} = \sum_{k=1}^{m} a_{ik}b_{kj}$$

Shorthand:

$$(A \cdot B)_{ij} = \sum_{k} a_{ik}b_{kj}$$

Transposing:

$$\left[(A \cdot B)^T\right]_{ij} = (A \cdot B)_{ji} = \sum_{k} A_{jk}B_{ik}$$

***Example** 6* **: Let X be a matrix for which X^{-1} exists. Prove that the inverse of X^T exists and is $(X^{-1})^T$**

$$\mathbb{1}^T = \mathbb{1}$$
$$\Rightarrow (X^{-1}X)^T = \mathbb{1}$$
$$\Leftrightarrow X^T(X^{-1})^T = \mathbb{1}$$

1.5. TENSORS

Vectors = tensors of rank 1, matrices = tensors of rank 2

The standart basis for $\mathbb{R}^{n \times m \times l}$ consists of $n \cdot m \cdot l$ basis vectors E_{ijk} where $1 \leq i \leq n, 1 \leq j \leq m, 1 \leq k \leq l$, such that all components of E_{ijk} are zero except at position i, j, k where the value of the component is 1.

2. FUNCTIONS OF SEVERAL VARIABLES

Let D and R be two sets. A mapping $f : D \to R$ that associates to each element $x \in D$ exactly one element of $f(x) \in R$ is called function.

Term	*Definition*
Domain of f	D
Range of f	R
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$

2.1. IMAGE CLASSIFICATION

A 128×256 pixel image I with 3 color channels (RGB) can be represented as a $128 \times 256 \times 3$ matrix (tensor of rank 3). The red value of a pixel at row 100, column 12 can be indexed with $I_{100,12,1}$. A function to determine whether the image was taken at day (0) or night (1) would have the following signature:

$$f : \mathbb{R}^{128 \times 256 \times 3} \to \{0, 1\}$$

2.1.1. Binary classification

TODO:

$$f : X \to \mathbb{R}, x \in X$$

$f(x)$ is called a one dimensional feature vector representing x. Afterwards $y = f(x)$ is mapped using a sigmoid function

$$\sigma : \mathbb{R} \to [0; 1]$$

which allows us to interpret the resulting value as a probability for x falling into the first category A. As soon as $\sigma(f(x))$ exceeds a certain threshold value τ, the system predicts x to fall into category A. Most of the time the logistic function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

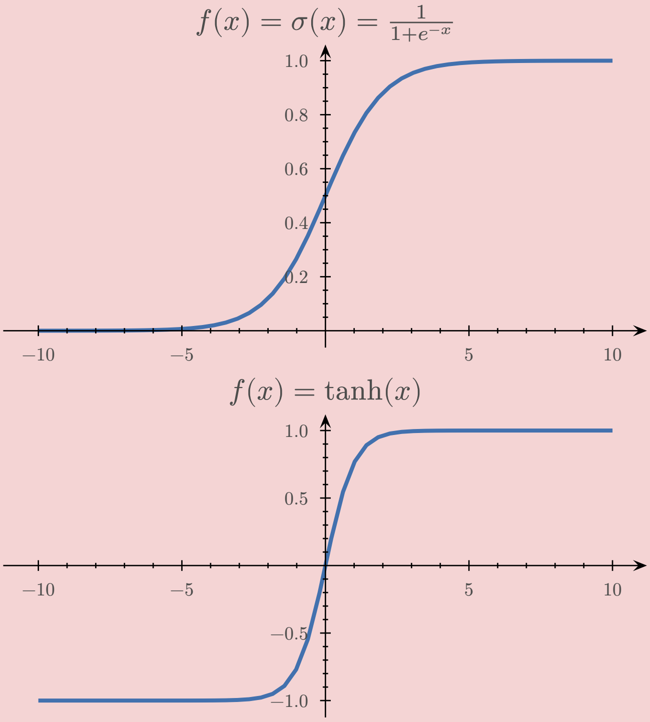
is used.

…

…

2.2. SIGMOID FUNCTION

TODO:



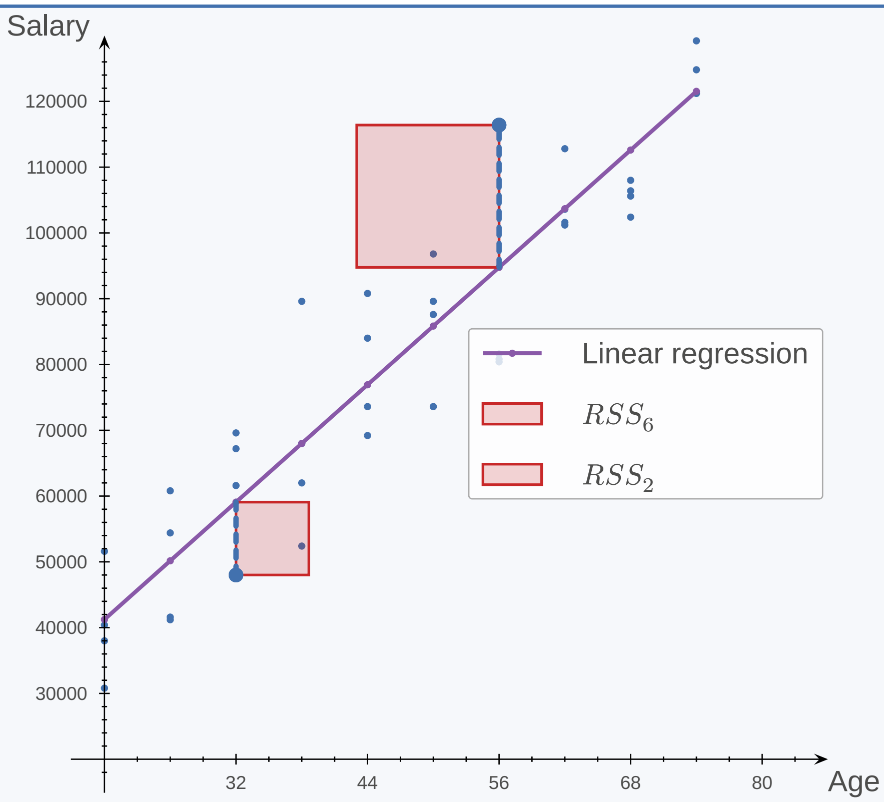
2.3. RESIDUAL SUM OF SQUARE

Residual sum of square (RSS) is a statistical method that helps identify the level of discrepancy in a dataset not predicted by a regression model. Thus, it measures the variance in the value of the observed data when compared to its predicted value as per the regression model.

$$RSS = \sum_{i=1}^{N} \underbrace{(y_i - f(x_i))^2}_{\substack{\text{Represented as red} \\ \text{squares in the example}}}$$

$$RSS(m, b) = \sum_{i=1}^{N} (y_i - (mx_i + b))^2 \geq 0, RSS : \mathbb{R}^2 \to \mathbb{R}$$

***Example** 7* **: Linear regression of salaries by age**



2.4. VISUALIZING FUNCTIONS

If $f : D \to R$ is a function from $D \subset \mathbb{R}^n$ to $R \subset \mathbb{R}^m$ its graph is defined as:

$$\operatorname{graph}(f) = \{(\vec{x}, \vec{y}) \in D \times R \mid x \in D \wedge \vec{y} = f(\vec{x})\} \subset D \times R \subset \mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$$

Due to the fact that the visual imagination of humans is limited to at most 3 dimensions, the graph of a function is a useful concept for graphical illustration if and only if $n + m \leq 3$.

***Example** 8*

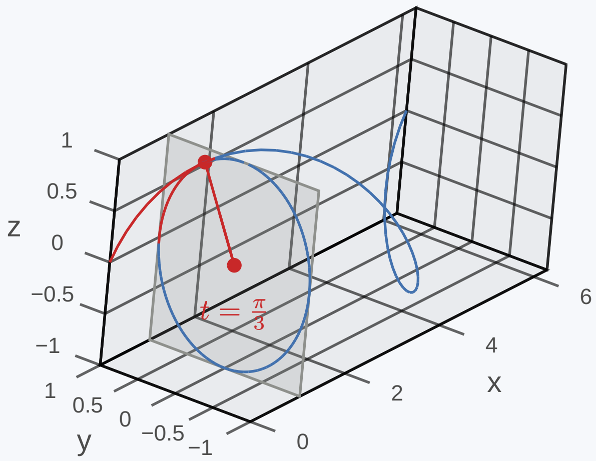
The graph of the function

$$f : \begin{cases} \mathbb{R} \to \mathbb{R}^2 \\ t \ \mapsto (\cos t, \sin t) \end{cases}$$

is

$$\begin{aligned} \operatorname{graph}(f) &= \left\{(t, x, y) \in \mathbb{R}^3 \;\middle|\; (x, y) = f(t) \wedge t \in \mathbb{R}\right\} \\ &= \left\{(t, \cos t, \sin t) \in \mathbb{R}^3 \;\middle|\; t \in \mathbb{R}\right\} \end{aligned}$$

It is displayed below and illustrates how f maps t to the corresponding values



***Example** 9*

The graph of the function

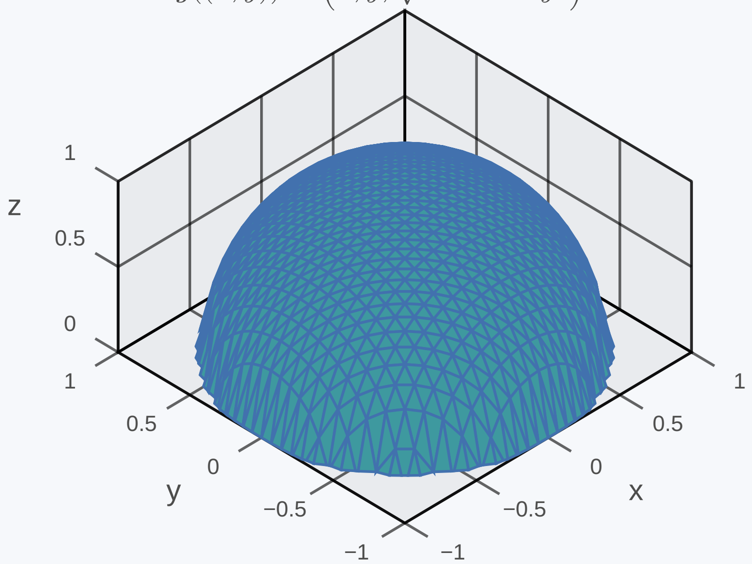
$$g : \begin{cases} \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\} & \to \mathbb{R}^2 \\ (x,y) & \mapsto \sqrt{1 - x^2 - y^2} \end{cases}$$

is

$$\begin{aligned} \operatorname{graph}(g) &= \left\{(x,y,z) \in \mathbb{R}^3 \;\middle|\; z = \sqrt{1-x^2-y^2} \wedge x^2+y^2 \leq 1\right\} \\ &= \left\{\left(x,y,\sqrt{1-x^2-y^2}\right) \in \mathbb{R}^3 \;\middle|\; x^2+y^2 \leq 1\right\} \end{aligned}$$

It is displayed below and illustrates how f maps (x,y) to the corresponding values

$$g((x,y)) = \left(x, y, \sqrt{1-x^2-y^2}\right)$$



2.4.1. Curves and surfaces

Term	*Definition*
Curves	Vector valued functions of one variable, i.e. they map one-dimensional input to multi-dimensional output
n-dimensional curve	A continuous function $f : I \to Z$ that maps an interval $I \subset \mathbb{R}$ into a subset $Z \subset \mathbb{R}^n$
Surfaces	Real valued functions of multiple variables, i.e. they map multi-dimensional input to one dimensional output
n-dimensional hyper-surface	A continuous real valued function $f : D \to Z$ that maps a sufficiently large subset $D \subset \mathbb{R}^n$ into a subset $Z \subset \mathbb{R}$ If $n = 2$ a hypersurface is also simply called surface. If $n = 1$ a hyper-surface is simply a continuous real valued function of one variable.

The graph of both, an n-dimensional curve and an n-dimensional hyper-surface, is a subset of $\mathbb{R}^{n+1}$. A graphical illustration of such a graph therefore requires $n \leq 2$. However, unlike surfaces which are typically illustrated as in Figure 1 , a curve $c : I \to \mathbb{R}^n$ is usually not illustrated through

$$\operatorname{graph} c = \{(t, y) \in R \times R^n \mid y = c(t) \land t \in I\}$$

Instead curves are typically visualized by skipping the independent variable t from the graph, i.e. by visualizing the image of the curve

$$c(I) = \{y \in R^n \mid y = c(t) \land t \in I\}$$

which means that we are projecting the graph of the curve onto the gray plane that is spanned from the dependent variables. As this kind of plot saves one dimension we can also visualize curves for which the range Z is a subset of $\mathbb{R}^3$.

***Example** 10*

The graph of the function

$$f : \begin{cases} \mathbb{R} \to \mathbb{R}^2 \\ t \mapsto (\cos t, \sin t) \end{cases}$$

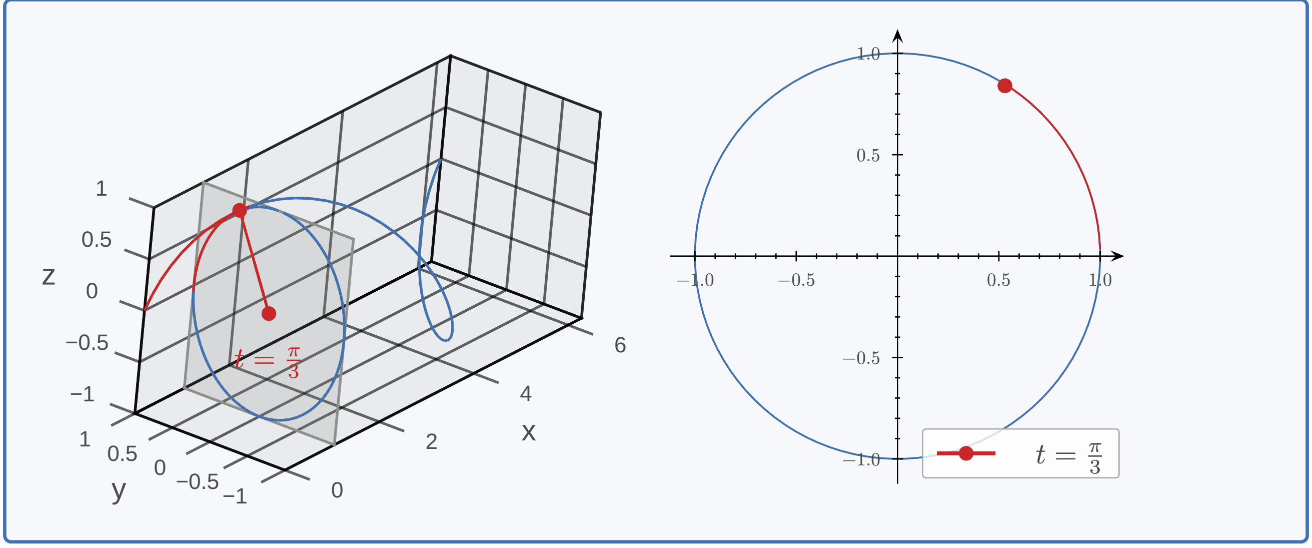
is

$$\begin{aligned} \operatorname{graph}(f) &= \left\{(t, x, y) \in \mathbb{R}^3 \;\middle|\; (x, y) = f(t) \land t \in \mathbb{R}\right\} \\ &= \left\{(t, \cos t, \sin t) \in \mathbb{R}^3 \;\middle|\; t \in \mathbb{R}\right\} \end{aligned}$$

and its image is the set

$$\begin{aligned} f(\mathbb{R}) &= \left\{f(t) \in \mathbb{R}^2 \;\middle|\; t \in \mathbb{R}\right\} \\ &= \left\{(\cos t, \sin t) \in \mathbb{R}^2 \;\middle|\; t \in \mathbb{R}\right\} \end{aligned}$$

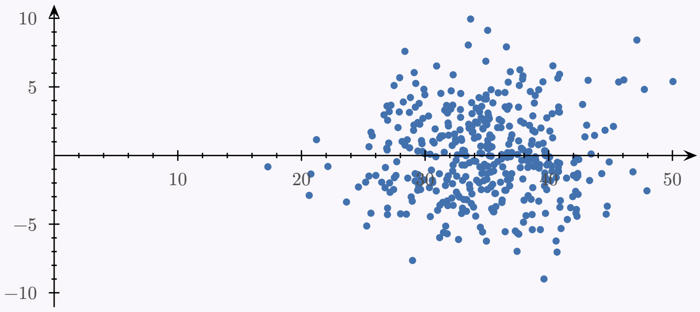
TODO:
3d axis order



Unlike for image processing, curves are of little importance in machine learning. On the contrary, hypersurfaces belong to the most important functions in machine learning. This is because, to a large extend, machine learning can be thought of being a branch of statistics. In statistics the most important class of functions are probability distributions, which are functions mapping subsets of R^n into R^+.

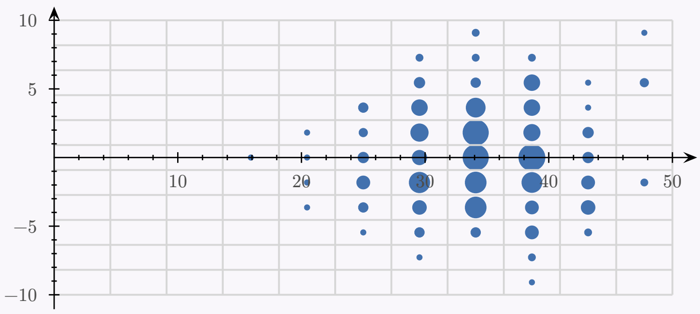
***Definition** 2* **:** **Histogram**

Consider having some amount of 2D datapoints:

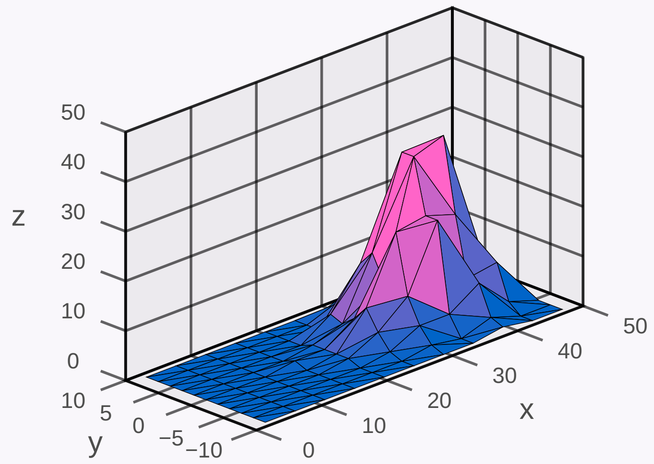


We can place a grid over the diagram and count, how many points are in each of the boxes.

TODO:
pt3d builtin



We can now plot a diagram that maps the x and y coordinate of the center of each of the boxes to the number of hits of this particular box and thus end up with a diagram that is called a ***histogram***.



We can try to create a histogram with boxes of infinitely small size. The problem with this approach is, that the number of hits per box decreases when the size of the boxes is getting smaller.

In order to compensate that, we therefore need more datapoints. If we do that, the histogram is being transformed into a function, that associates a number to each position on the diagram, that can be interpreted as likelihood for a point to be almost next to this position: the higher this number is, the more likely it is. This function is called the **probability density** of the experiment.

2.4.1.1. Vector similarity

In addition to probability theory, geometric concepts are used in machine learning. One approach to decide, whether two images are similar or not, is to first extract features (such as corner-points, edges, etc.) from each of the images, and store these features in a list of numbers, which is called ***feature vector***. It is reasonable to assume, that similar images are mapped to similar feature vectors.

Two vectors v and w are **similar**, if both vectors head into the same direction and are of similar length, so if the difference $v - w$ between both vectors has a small length. In vector geometry, the length of a vector is called a **norm**, and the distance between two vectors is called a **metric**. A norm $\|\cdot\|$ is thus a function that associates with a vector x a positive number $\|x\|$, which is interpreted as length of x.

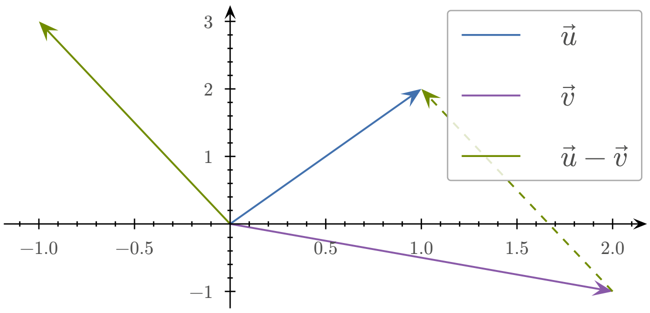
$$\|\cdot\| : \begin{cases} \mathbb{R}^n & \to \mathbb{R}^+ \\ x & \mapsto \sqrt{\sum_{i=1}^n x_i^2} \end{cases}$$

$=$ n-dimensional Euclidean norm or l^2-norm, written also as $\|x\|_2$.

Once a vector space is equipped with a norm $\|.\|$, we can also associate a metric with that vector space, which uses the notion of a length to measure the distance between pairs of vectors:

$$d(\cdot,\cdot) : \begin{cases} \mathbb{R}^n \times \mathbb{R}^n & \to \mathbb{R}^+ \\ (u,v) & \mapsto \|u - v\| \end{cases}$$

$=$ metric. The value $d(u,v)$ is identical to the length of the vector that heads from v to u.



2.4.2. Vector valued functions of several variables

TODO:
function signature

In machine learning many functions are vector valued functions of many variables. A very important example that falls into this category are linear transformations. Neural networks are typically defined to be a stack of unknown linear transformations, each of which being followed by a rather simple, predefined non-linear operation. Training a neural network is therefore almost equivalent to finding suitable linear transformations that make the network perform the desired operation.

Besides curves and surfaces there are functions that take multi-dimensional input and deliver multi-dimensional output. In order to get an idea how these functions look like, we often display low-dimensional "sections" of the function that can either be displayed as curves or as surfaces. There is however another way, how these functions can be displayed, namely using a **vector field**.

***Example** 11* **: Spaceship (vector field)**

A space-ship traveling close to the sun is experiencing a gravitational force that - outside the sun - can be described using the following function:

$$F : \begin{cases} \{r \in \mathbb{R}^3 \mid |r| > r_{\text{Sun}}\} \to \mathbb{R}^3 \\ r \qquad\qquad\qquad\qquad\quad \mapsto -G \cdot m_{\text{Sun}} m_{\text{Ship}} \frac{r}{\|r\|^3} \end{cases}$$

r_{Sun} = radius of the sun, G = gravitational constant, $m_{\text{Sun}}, m_{\text{Ship}}$ = masses of the sun and space ship.

If the center of the sun is located at $r = 0$, the input vector r denotes the position of the space ship as seen from the center of the sun and F denotes the gravitational force that the space ship is experiencing.

From a mathematical perspective the constants are not important. Therefore we simplify the formula by setting all of the constants to 1, which leads to:

$$F : \begin{cases} \{r \in \mathbb{R}^3 \mid |r| > 1\} \to \mathbb{R}^3 \\ r \qquad\qquad\qquad\quad \mapsto -\frac{r}{\|r\|^3} \end{cases}$$

In order to better understand the formula of the force field, we calculate the force experienced from the space ship at position $r = (1, 2, 2)^T$. Note first, that r is in the domain of definition of F , because its norm is greater than $r_{\text{Sun}} = 1$

$$\|r\| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$$

This simply indicates, that the spaceship resides at a location exterior to the sun. We can therefore use this to calculate the force that the space ship is experiencing:

$$F\begin{pmatrix}1\\2\\2\end{pmatrix} = -\frac{1}{\|(1,2,2)^T\|^3}\begin{pmatrix}1\\2\\2\end{pmatrix} = -\frac{1}{3^3}\begin{pmatrix}1\\2\\2\end{pmatrix} = -\frac{1}{27}\begin{pmatrix}1\\2\\2\end{pmatrix}$$

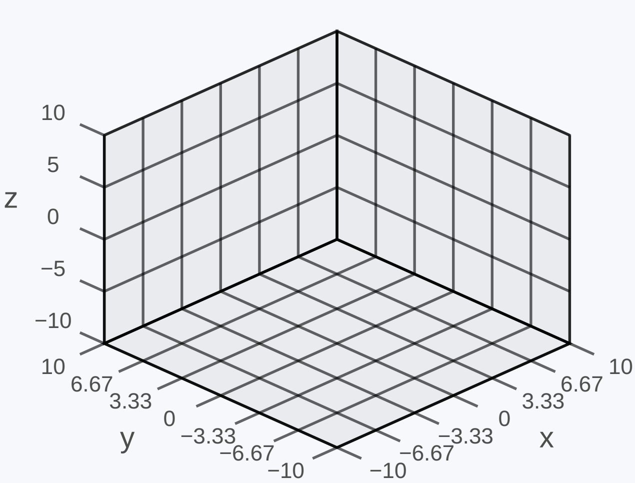
We can now continue to calculate values of the force F at other locations and try to graphically visualize the result. Since the graph of F

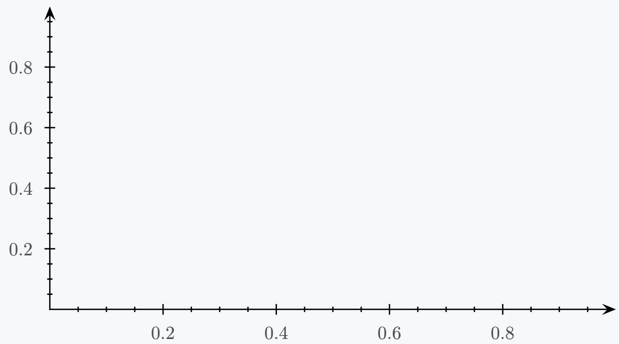
$$\operatorname{graph}(F) = \left\{(r,f) \in \mathbb{R}^3 \times \mathbb{R}^3 \;\middle|\; \|r\| > 1 \land f = -\frac{r}{\|r\|^3}\right\}$$

is a subset of a $3+3=6$-dimensional space, we cannot however not draw the graph of F directly in any meaningful way. A possibility to still visualize this function, is to draw the function as **vector field**. By this we mean that we place 3-dimensional arrows representing F at different points in a 3-dimensional space that identify the physical location of the space ship. The arrow that is placed at position r has a length proportional to the norm of the force $F(r)$ and a direction parallel to the direction of the force. It thus tells you the strength and direction in which the space ship is pulled due to the gravitational force.

TODO:
pt3d vector field

TODO:
2d repr





The mathematical tool for determining the “strength of the field” is the **norm of a vector**, which measures the length of a vector and thus encodes the strength of a field. If we calculate $\|F(r)\|$ in our case, we get

$$\|F(r)\| = \left\|-\frac{r}{\|r\|^3}\right\| = \frac{\|r\|}{\|r\|^3} = \frac{1}{\|r\|^2}$$

which illustrates that the force is getting smaller, the larger $|r|$ is

The mathematical tool for determining the “direction of a field” are **unit vectors**. The unit vector of r is a vector of length one, that points into the same direction as r. It thus encodes information regarding the direction of r, but not regarding its length. The unit vector of r is usually denoted by e_r. It can be calculated via

$$e_r = \frac{1}{\|r\|}\cdot r$$

2.5. CALCULUS OF CURVES

TODO:
function signature

Definition 3 : Coordinate functions

Let $I \subset \mathbb{R}$ be an interval and $c : I \to \mathbb{R}^n$ be an arbitrary curve. Then there are n (contiguous) coordinate functions $c_i : I \to \mathbb{R}$, such that

$$c(t) = \begin{pmatrix} c_1(t) \\ c_2(t) \\ \vdots \\ c_n(t) \end{pmatrix}$$

and vice versa: If n real valued (continuous) functions $c_i : I \to \mathbb{R}$ with a common interval I as domain of definition are given, we can define a curve $c(t)$ whose coordinate functions are $c_1, c_2, ..., c_n$.

The decomposition of curves into coordinate functions allows us to generalize most of the concepts from previous analysis courses to curves:

We consider the curve

$$f : \begin{cases} (0; \infty) \to \mathbb{R}^2 \\ t \qquad\quad \mapsto \begin{pmatrix} \cos(t) \\ \frac{\sin(t)}{t} \end{pmatrix} \end{cases}$$

The component functions are

$$f_1(t) : \cos(t)$$
$$f_2(t) : \frac{\sin(t)}{t}$$

We can thus argue:

Definition 4 : Continuity

Since both of the component functions are continuous, f is continuous.

Definition 5 : Limit point

$$\lim_{t\to 0} f_1(t) = \lim_{t\to 0} \cos(t) = \cos(0) = 1$$
$$\lim_{t\to 0} f_2(t) = \lim_{t\to 0} \frac{\sin(t)}{t} \overset{\text{"0/0"}}{=} \lim_{t\to 0} \frac{\frac{\mathrm{d}}{\mathrm{d}t}\sin(t)}{\frac{\mathrm{d}}{\mathrm{d}t}t} = \lim_{t\to 0} \frac{\cos(t)}{1} = \cos(0) = 1$$
$$\Rightarrow \lim_{t\to 0} f(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

***Definition** 6* : **Derivative**

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}f_1(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\cos(t) = -\sin(t)\\ \frac{\mathrm{d}}{\mathrm{d}t}f_2(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\sin(t)}{t} = \frac{\cos(t)\cdot t - \sin(t)\cdot 1}{t^2} = \frac{\cos(t)}{t} - \frac{\sin(t)}{t^2}\\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}f(t) &= \begin{pmatrix} -\sin(t)\\ \frac{\cos(t)}{t} - \frac{\sin(t)}{t^2}\end{pmatrix}\end{aligned}$$

***Definition** 7* : **Linearisation at $t = \pi$**

$$\begin{aligned}f_1(t) &\approx f_1(\pi) + f_1'(\pi)(t-\pi) = \cos(\pi) - \sin(\pi)(t-\pi) = -1\\ f_2(t) &\approx f_2(\pi) + f_2'(\pi)(t-\pi) = \frac{\sin(\pi)}{\pi} + \left(\frac{\cos(\pi)}{\pi} - \frac{\sin(\pi)}{\pi^2}\right)(t-\pi) = -\frac{1}{\pi}(t-\pi)\\ \Rightarrow f(t) &\approx f(\pi) + f'(\pi)(x-\pi) = \begin{pmatrix}-1\\ -\frac{1}{\pi}(t-\pi)\end{pmatrix} = \begin{pmatrix}-1\\ 0\end{pmatrix} - (t-\pi)\begin{pmatrix}0\\ \frac{1}{\pi}\end{pmatrix}\end{aligned}$$

TODO:
diagram

For any $\vec{a}, \vec{v} \in \mathbb{R}^n$ a curve of the form $g: \begin{cases}\mathbb{R}\to\mathbb{R}^n\\ t\mapsto \vec{a}+t\cdot\vec{v}\end{cases}$ corresponds to a line going through the point a in the direction of v.

TODO:
Notes 30/31

***Definition** 8* : **Reparametrization**

Let

$$c : \begin{cases} I \to \mathbb{R}^n\\ t \mapsto c(t)\end{cases}$$

be a curve and $h: J \to I$ an arbitrary bijective function. Then the image of the curve

$$d : \begin{cases} J \to \mathbb{R}^n\\ s \mapsto c(h(s))\end{cases}$$

is identical to the image of the curve $c(t)$:

$$c(I) = d(J)$$

We call $d(s)$ ***reparametrization*** of $c(t)$ using $t = h(s)$

TODO:
rest of chapter

***Definition** 9* : **Image of a curve**

The image of a curve

$$c : \begin{cases} I \to \mathbb{R}^n \\ t \mapsto c(t) \end{cases}$$

can be envisioned as a road that is passed from a passenger, who reaches position $c(t)$ at time t. Thus:

- The **derivative** $c'(t)$ is a **measure of the speed** of the person at time t.
- The **derivative** $c'(t)$ points into a **direction tangent to the road** at position $c(t)$
- The **norm** $\|c'(t)\|$ of the derivative is **a measure of the speed** with which the person is moving **at time** t.

If we reparametrize the curve c with a bijective function $t = h(s)$ the image of the curve

$$d(s) = c(h(s))$$

is identical to the image of c and thus corresponds to the same road. It is however passed by another person, who is passing point $d(s) = c(h(s))$ with a velocity

$$d'(s) = c'(h(s)) \cdot h'(s)$$

that is $h'(s)$ faster (or slower) than the velocity of the first person at time $t = h(s)$.

TODO:
formal proof (Notes 33/34)

2.6. CALCULUS OF REAL VALUED FUNCTIONS IN MANY VARIABLES

TODO:
function signature

2.6.1. Partial derivative and gradient

Let $f : D \to \mathbb{R}$ be an arbitrary real valued function and $x \in D \subset \mathbb{R}^n$ be an arbitrary vector in the domain of f. We consider the domain D of f to be sufficiently large for doing calculus at x, if for every vector $v \in \mathbb{R}^n$ there is a small interval $(-\varepsilon; \varepsilon), \varepsilon > 0$, such that for all $t \in (-\varepsilon; \varepsilon)$ the vector $x + t \cdot v \in D$. This condition guarantees that any curve that hits the domain of f, remains there for at least a short period of time, and is one of the properties that needs to be satisfied in order to denote f as a hyper-surface. The other property that is required for f to be a hyper-surface, is that f is continuous at any point of its domain.

***Definition** 10* : **Partial derivative**

The ***partial derivative*** of f with respect to the i-th coordinate function is defined through

$$\frac{\partial}{\partial x_i} f(x) = \lim_{t \to 0} \frac{f(x + t \cdot e_i) - f(x)}{t}$$

***Definition** 11* : **Gradient**

The ***gradient*** of f is an n-dimensional row vector, whose components correspond to the various **partial derivatives**.

$$\nabla f(x) = \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), ..., \frac{\partial}{\partial x_n} f(x)\right)$$

***Example** 12* **:** $f : \begin{cases} \mathbb{R} \times \mathbb{R} \to \mathbb{R} \\ (x, y) \ \mapsto x^2 y^3 \end{cases}$

$$\begin{aligned} f'_x(x, y) &= \frac{\partial f}{\partial x} = 2x \cdot y^3 \\ f'_y(x, y) &= \frac{\partial f}{\partial y} = x^2 \cdot 3y^2 \\ \nabla f(x, y) &= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x \cdot y^3, x^2 \cdot 3y^2) \end{aligned}$$

2.6.2. Compositions of functions

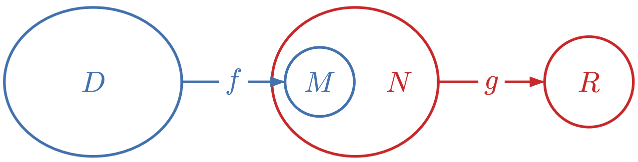
We can use the notion of dependent variables as an alternative representation of functions, this notion allows us to define functions without explicitly stating its arguments.

TODO:
example (Notes 40)

2.6.3. Partial and total derivative

TODO:

2.6.4. Commutative diagrams



$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & M \arrow[r, "g"] & R \end{tikzcd}$$

Let us assume that

$$f : \begin{cases} D \to M \\ x \ \mapsto f(x) \end{cases}, \quad g : \begin{cases} N \to R \\ y \ \mapsto g(y) \end{cases}, \quad h : \begin{cases} D \to R \\ x \ \mapsto g(f(x)) \end{cases} = g \circ f$$

The graph

$$D \xrightarrow{\quad f \quad} M \subset N \xrightarrow{\quad g \quad} R$$

illustrates that f is a function from D to M, g is a function from N to R and M is a subset of N. Alternatively, for every set Q, such that $M \subset Q \subset R$ we could also have written

$$D \xrightarrow{\quad f \quad} Q \xrightarrow{\quad g \quad} R$$

indicating that f maps D into a subset of Q, which in turn is a subset of the domain of g and thus can be mapped to R using g.

A ***commutative diagram*** is an extension of the previous diagram, in which we add an additional arrow pointing from D to R.

$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "h"] & N \arrow[r, "g"] & R \end{tikzcd}$$

By labeling this additional arrow with h, a commutative diagram can be used to define a function from D to R, that is behaving such, that it maps every $x \in D$ to exactly the same element as the composition of g with f.

***Example** 13* **:**

$$\begin{aligned} g(x) &= x^2 \\ f(x) &= \sqrt{x} \\ f(g(x)) &= \sqrt{x^2} &= |x| \\ g(f(x)) &= \left(\sqrt{x}\right)^2 &= |x| \end{aligned}$$

$$\begin{tikzcd} \mathbb{R}^+ \arrow[r, "\cdot^2"] \arrow[d, "\sqrt{\cdot}"'] \arrow[dr, dotted, "|\cdot|"] & \mathbb{R}^+ \arrow[d, "\sqrt{\cdot}"] \\ \mathbb{R}^+ \arrow[r, "\cdot^2"'] & \mathbb{R}^+ \end{tikzcd}$$

2.7. CALCULUS OF VECTOR VALUED FUNCTIONS OF MANY VARIABLES

TODO:
function signature

2.7.1. Jacobian matrix and linearisation

In the theory of real valued functions in one variable the linearisation of $f : \mathbb{R} \to \mathbb{R}$ at $x_0 \in \mathbb{R}$ is done via

$$f(x) \approx f(x_0) + f'(x_0) \cdot (x - x_0), x \approx x_0$$

and can be interpreted as:

1) First order Taylor polynomial of f, which implies that the linearisation of f at x_0 is that first order polynomial which best approximates f in the vicinity of x_0.
2) Identifying that particular line that fits best to the graph of f in a neighborhood of x_0. This line is known as tangent of f at x_0.
3) Definition of the derivative of f at x_0.

$$f'(x_0) \approx \frac{f(x) - f(x_0)}{x - x_0} \text{ for } x \approx x_0, x \neq x_0$$

   tells us that both sides of this equation become equal for $x \to x_0$, and therefore equivalent to the definition of the derivative of f at x_0

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

In the theory of vector valued functions of many variables, the linearisation of a vector valued function in n variables $f : \mathbb{R}^n \to \mathbb{R}^m$, itself has to be a mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$. Further, the right hand side has to be an affine transformation, i.e. a function of the form

$$A_{a,M}(x) = a + M \cdot x$$

in which M is a matrix and a is a vector. Since we expect that $A_{a,M}$ to map $\mathbb{R}^n$ to $\mathbb{R}^m$, we can infer that $a \in \mathbb{R}^m$ and $M \in \mathbb{R}^{m \times n}$.

The **derivative of a vector valued function** $f : \mathbb{R}^n \to \mathbb{R}^m$ can thus be expected to be a matrix $M \in \mathbb{R}^{m \times n}$. This matrix is called ***Jacobian matrix*** and it can be shown that this matrix is comprised of all partial derivatives of all component functions of f.

***Definition** 12 :* **Jacobian matrix**

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a some vector valued function and $x_0 \in \mathbb{R}^n$ be a point, where the partial derivatives of all component functions $\partial_{x_k} f_l(x_0)$ exist. The $(m \times n)$-matrix that aggregates all partial derivatives of f at x_0

$$J_f(x_0) = \begin{pmatrix} \partial_{x_1} f_1(x_0) & \partial_{x_2} f_1(x_0) & \dots & \partial_{x_n} f_1(x_0) \\ \partial_{x_1} f_2(x_0) & \partial_{x_2} f_2(x_0) & \dots & \partial_{x_n} f_2(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_{x_1} f_m(x_0) & \partial_{x_2} f_m(x_0) & \dots & \partial_{x_n} f_m(x_0) \end{pmatrix}$$

such that the l-th row of $J_f(x_0)$ corresponds to the gradient of the l-th coordinate function f_l, is called ***Jacobian matrix*** of f at x_0. Besides of the symbol $J_f(x_0)$ the following notations are also used for the Jacobin matrix at x_0:

$$f'(x_0) = \left.\frac{\partial f}{\partial x}\right|_{x=x_0} = J_f(x_0)$$

***Example** 14 :* **Jacobian matrix**

TODO:

***Definition** 13 :* **Linearisation**

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a some vector valued function and $x_0 \in \mathbb{R}^n$ be an arbitrary point from the domain of definition of f If $J_f(x_0) = \left.\frac{\partial f}{\partial x}\right|_{x=x_0}$ is the Jacobin matrix of f at x_0, then

$$f(x) \approx f(x_0) + J_f(x_0) \cdot (x - x_0) \text{ for } x \approx x_0$$

The function on the right-hand side is called ***linearisation*** of f at x_0.

***Example** 15 :* **Linearisation**

TODO:

2.8. SUMMARY DERIVATIVES

Function	*Derivative*
$f: \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{\mathrm{d}}{\mathrm{d}x} f_1(x) \\ \frac{\mathrm{d}}{\mathrm{d}x} f_2(x) \\ \vdots \\ \frac{\mathrm{d}}{\mathrm{d}x} f_n(x) \end{pmatrix}$
$f: \begin{cases} \mathbb{R}^n \to \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \to \mathbb{R}^{1\times n} \\ x \mapsto y \end{cases} = \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x)\right) = \nabla f$
$f: \begin{cases} \mathbb{R}^n \to \mathbb{R}^m \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \to \mathbb{R}^{m\times n} \\ x \mapsto y \end{cases} = \begin{pmatrix} \nabla f_1(x) \\ \nabla f_2(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \frac{\partial}{\partial x_2} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \frac{\partial}{\partial x_1} f_2(x) & \frac{\partial}{\partial x_2} f_2(x) & \dots & \frac{\partial}{\partial x_n} f_2(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \frac{\partial}{\partial x_2} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix} = J_f$

2.8.1. Chain-rule for vector valued functions in many variables

***Definition** 14* **: Jacobian matrix chain rule**

Let f and g be given such that the following diagram commutes

$$\begin{tikzcd} \mathbb{R}^n \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & \mathbb{R}^k \arrow[r, "g"] & \mathbb{R}^m \end{tikzcd}$$

Then we can calculate the Jacobian matrix of $g \circ f$ using the chain rule

$$\frac{\partial g(f(x))}{\partial x} = \left.\frac{\partial g(f)}{\partial f}\right|_{f=f(x)} \cdot \frac{\partial f(x)}{\partial x} = J_g(f(x)) \cdot J_f(x)$$

***Definition** 15* **: Chain rule for hyper-surfaces**

Let $f : \mathbb{R}^m \to \mathbb{R}$ be a hyper-surface and $g : \mathbb{R}^n \to \mathbb{R}^m$ a vector valued function, such that $h = f \circ g$ is defined. Then

$$\nabla h(x) = \nabla f(g(x)) = \nabla f(g)|_{g=g(x)} \cdot \frac{\partial}{\partial x} g(x)$$

***Definition** 16* **: Properties of differentiation of hyper-surfaces**

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ be hyper-surfaces. Then

$$\begin{aligned}
\nabla(f+g) &= \nabla f + \nabla g \\
\nabla(f-g) &= \nabla f - \nabla g \\
\nabla(f \cdot g) &= g\nabla f + f\nabla g \\
\nabla \frac{f}{g} &= \frac{g\nabla f - f\nabla g}{g^2}
\end{aligned}$$

TODO:
notes 26.03.26

TODO:
BIAS: We say that a model has low bias if it leads to a good approximation of reality, and that a model has high bias if it cannot accurately reflect our measurements.

NOISE: Typically the data in the training dataset is the outcome of some measurement and thus contains noise. “Noise” has a negative effect during training, and in general we do not want the system to “learn noise”. One way to prevent a machine learning system from learning “noise” is to use large datasets, as statistics tells us that the average amount of noise tends to zero for large datasets.

COST FUNCTION: To measure the deviation between prediction and expectation we use cost functions. Technically speaking, a cost function is a real valued function that maps model parameters to $\mathbb{R}^+$ in such a way that parameter choices resulting in lower bias correspond to lower values of the cost function (low bias = low cost). One way to define a cost function, is to use the Euclidean distance between the (column) vector of predicted results and the (column) vector of expected results. We can do this by first adding a column containing the values of the predicted results, which depends on the model parameters and a column that measures the deviation of the prediction from the expectation, which is called the residuum r:

3. GEOMETRY OF HYPER-SURFACES

3.1. INTRODUCTION

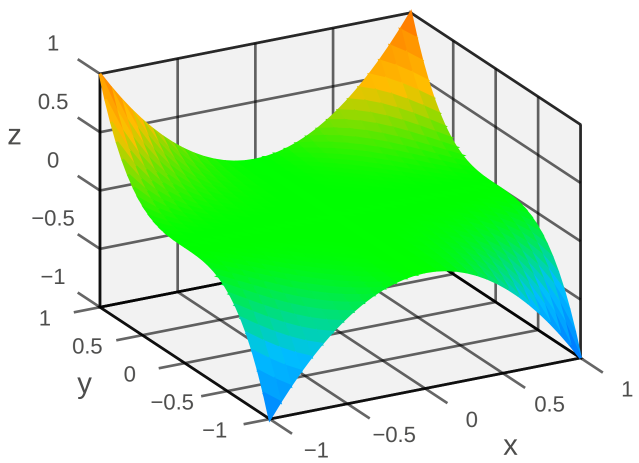
Term	***Definition***
Training sample	Knowledge of in- and outputs prior to learning
Supervised learning	A process for finding suitable values of parameters that make the system respond as desired for as many training samples as possible
Cost functions	Process of finding the values in supervised learning. Function of n parameters that maps them to a high value if the misclassification is high and low if otherwise: $$c : \mathbb{R}^n \to \mathbb{R}$$
Train	Feed the system with training samples and find the best parameters in the process in searching for the (global) minimum of the cost function

3.1.1. Visualizing surfaces

To a large extend, the world we live in can be envisioned as a two dimensional surface that is embedded in three dimensional space. Using this analogy you can take away two important methods for visualizing surfaces: surface-plots (3d) and contour-plots (2d).

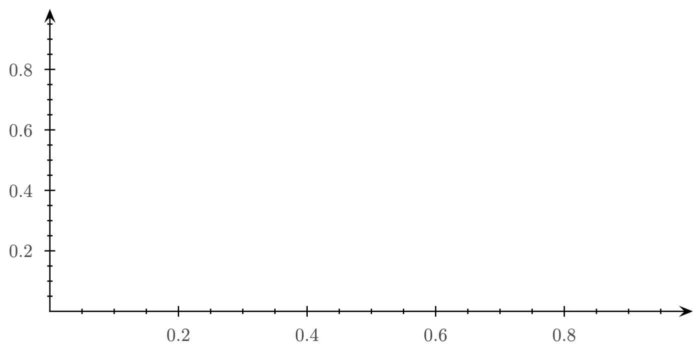
$$f(x,y) = x^2 y^3$$

Surface-plot:



Contour-plot:

TODO:



3.1.2. Tangent space and linearisation of a surface

***Definition** 17* **:** **Set of all tangent directions**

Let $f: \mathbb{R}^2 \to \mathbb{R}$ be a surface and $p = (x_0, y_0, f(x_0, y_0)) \in \operatorname{graph} f$. Then the set of all tangent directions to curves $c(t)$ with $c(t) = p$ that reside entirely inside graph f is given by

$$T = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ \partial_x f(x_0, y_0) \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ \partial_y f(x_0, y_0) \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

***Definition** 18* **:** **Tangent space**

The set of all points

$$T = \left\{ \begin{pmatrix} x_0 \\ y_0 \\ f(x_0, y_0) \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ \partial_x f(x_0, y_0) \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ \partial_y f(x_0, y_0) \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

that can be reached from p by traveling into a tangent direction, is a 2-dimensional plane, which is called ***tangent space*** of f at p.

***Definition** 19* **:** **Graph of a linearisation**

Let $f: \mathbb{R}^n \to \mathbb{R}$ be a real valued function in n variables and $x_0 \in \mathbb{R}^n$. Then the graph of the linearisation

$$l(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0)$$

is identical to the tangent space T_{p_0} of f at $p_0 = \left(((x_0))_1, ((x_0))_2, ..., ((x_0))_n, f(x_0)\right)^T$.

3.2. OPTIMIZATION PROBLEMS

3.2.1. Extreme values and stationary points

Let $f : D \to \mathbb{R}$ with $D \subset \mathbb{R}^n$ be a hyper-surface.

Term	***Definition***
Upper bound	Any number $u \in \mathbb{R}$, such that $f(x) \leq u$ for all $x \in D$ is called an ***upper bound*** of f
Lower bound	Any number $l \in \mathbb{R}$, such that $f(x) \geq l$ for all $x \in D$ is called a ***lower bound*** of f
Global maximum	An upper bound u is called ***global maximum value*** or ***absolute maximum value*** of f, if $u \in f(D)$. The value x_0 is called the ***global maximum point*** or ***absolute maximum point***
Global minimum	A lower bound l is called ***global minimum value*** or ***absolute minimum value*** of f, if $l \in f(D)$. The value x_0 is called the ***global minimum point*** or ***absolute minimum point***
Local maximum	Any point $x_0 \in D$ such that, whenever $x \approx x_0 \land f(x) \leq f(x_0)$ is called ***local maximum point*** and the value $f(x_0)$ is called ***local maximum value***
Local minimum	Any point $x_0 \in D$ such that, whenever $x \approx x_0 \land f(x) \geq f(x_0)$ is called ***local minimum point*** and the value $f(x_0)$ is called ***local minimum value***

***Definition** 20* **: Stationary**

Let $f : D \to \mathbb{R}$ with $D \subset \mathbb{R}^n$ be a hyper-surface that can be differentiated. Any point $x_0 \in D$ with $\nabla f(x_0) = 0$ is called ***stationary***.

***Observations** 3*

3.1. If x_0 is a local extremal point, then f is stationary at x_0, i.e. $\nabla f(x_0) = 0$.

Stationary points can also identify positions, at which a function is becoming flat into one direction without loosing its monotony.

Finally, there is a third scenario, which can occur: saddle points. Saddle points require functions that depend on at least two variables, because the domain of such functions is sufficiently large to move into at least two different directions without leaving the graph. In such a case it can happen, that a stationary point appears to be as a maximum value in one direction, and as a minimum value in another direction.

TODO:
diagrams (script 90) + head of page 91

3.2.2. Necessary and sufficient conditions for extremal values

Notation for second derivative:

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right) = \frac{\partial^2 f(x,y)}{\partial x^2} = \partial_{xx} f(x,y)$$

***Definition** 21* **: Hessian matrix**

Let $f : D \to \mathbb{R}$ with $D \subset \mathbb{R}^n$ be a hyper-surface. The matrix $H(x)$, whose components are

$$(H(x))_{ij} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f(x)$$

is called ***Hessian matrix***. The Hessian matrix is the generalization of the second order derivation to hyper-surfaces.

Let $z = f(c(t))$, c a curve passing our stationary point $c_0 = (x_0, y_0)$ at $t = t_0$ and

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T} \cdot \underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0) \end{pmatrix}}_{H} \cdot \underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0) \end{pmatrix}}_{v}$$

then
1) If you can show that $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v > 0)$, then $\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} > 0$ defines a ***local minimum***
2) If you can show that $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v < 0)$, then $\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} < 0$ defines a ***local maximum***
3) In both of those cases, i.e. $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v < 0 \vee v^T H v > 0)$, the point $c_0 = (x_0, y_0)$ defines a ***saddle point***
4) In all other cases it defines nothing and further investigation is needed

***Observations** 4*

4.1. The Hessian matrix is always symmetric, i.e. that $\partial_i \partial_j f = \partial_j \partial_i f$